

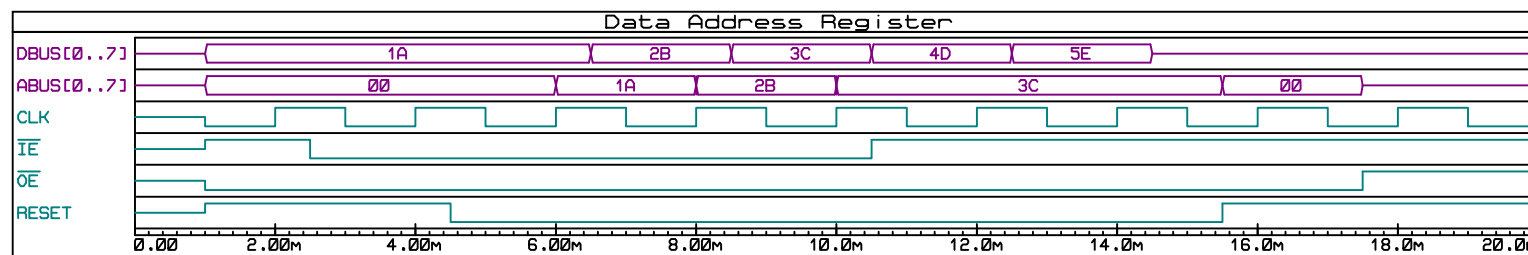
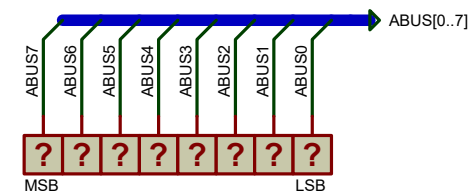
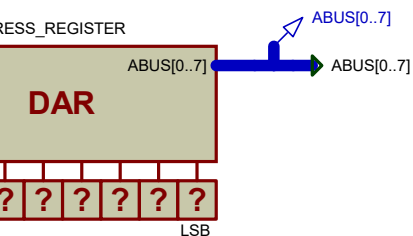
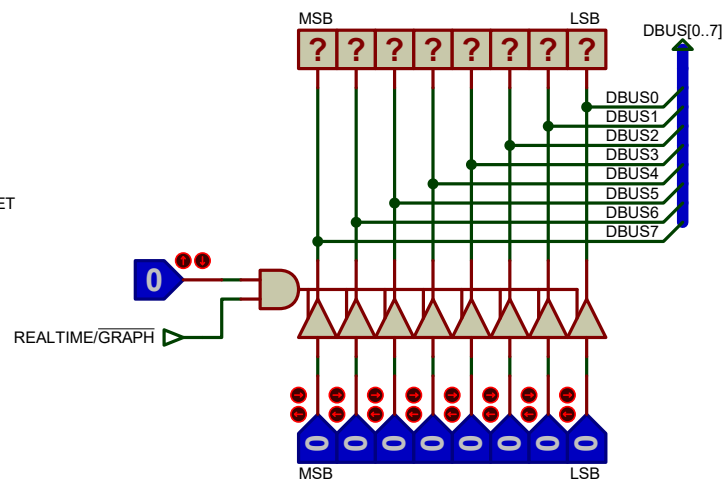
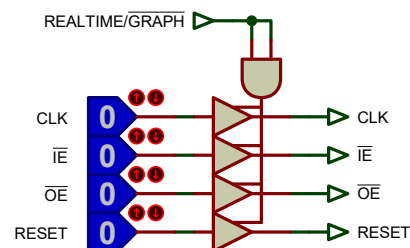
ROOT

The diagram illustrates the internal structure and signal flow of the Data Address Register (DAR). It includes the following components and connections:

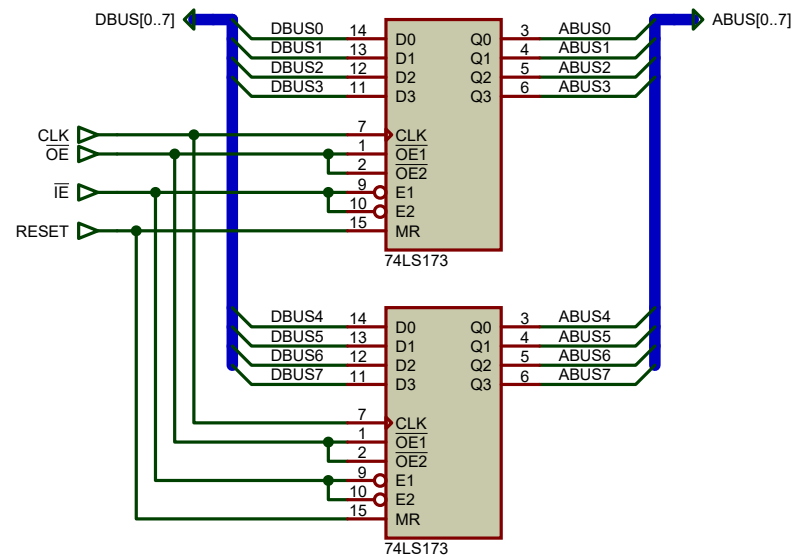
- TEST_DATA_ADDRESS_REGISTER:** A control register that receives the **REALTIME/GRAPH** signal. It has an **ON/OFF** control and outputs **DBUS[0..7]**, **CLK**, **IE**, **OE**, and **RESET** signals.
- DATA_ADDRESS_REGISTER (DAR):** The main register that receives **DBUS[0..7]** and **CLK** signals. It has control inputs for **IE**, **OE**, and **RESET**. It outputs **ABUS[0..7]** and has an 8-bit data bus output represented by boxes labeled **MSB** to **LSB**.
- Signal Processing:** The **REALTIME/GRAPH** signal is processed through a series of logic gates (AND, OR, NOT) to generate the **CLK**, **IE**, **OE**, and **RESET** signals for the registers.
- DBUS[0..7] and ABUS[0..7]:** The data buses are shown as 8-bit signals. The **DBUS[0..7]** is connected to the **DATA_ADDRESS_REGISTER**, and the **ABUS[0..7]** is connected to the **DATA_ADDRESS_REGISTER** output.
- Internal Connections:** The diagram shows the internal connections between the **TEST_DATA_ADDRESS_REGISTER** and the **DATA_ADDRESS_REGISTER**, including the **DBUS[0..7]** and **ABUS[0..7]** signals.

The timing diagram shows the signals for the Data Address Register over a period of 20.00m. The signals are:

- DBUS[0..7]:** A signal that changes from 1A to 2B at 6.00m, then to 3C at 8.00m, then to 4D at 10.00m, and finally to 5E at 12.00m.
- ABUS[0..7]:** A signal that changes from 00 to 1A at 6.00m, then to 2B at 8.00m, then to 3C at 10.00m, and finally to 00 at 12.00m.
- CLK:** A periodic clock signal with a period of 2.00m.
- IE:** An input enable signal that is active (low) from 2.00m to 10.00m.
- OE:** An output enable signal that is active (low) from 10.00m to 18.00m.
- RESET:** A reset signal that is active (low) from 4.00m to 16.00m.



DATA_ADDRESS_REGISTER



TEST_DATA_ADDRESS_REGISTER

*SCRIPT GENERATOR TEST_DAR

ALIAS ONOFF = \$ON\$/OFF
 ALIAS IE = \$IE\$
 ALIAS OE = \$OE\$

PIN ONOFF
 PIN DBUS[0..7] = FLT
 PIN IE = FLT
 PIN OE = FLT
 PIN RESET = FLT

IF ONOFF = TRUE THEN
 DBUS[0..7] = FLT
 IE = FLT
 OE = FLT
 RESET = FLT
ELSE
 DBUS[0..7] = \$1A AFTER 1m
 DBUS[0..7] = \$2B AFTER 6.5m
 DBUS[0..7] = \$3C AFTER 8.5m
 DBUS[0..7] = \$4D AFTER 10.5m
 DBUS[0..7] = \$5E AFTER 12.5m
 IE = 1 AFTER 1m
 IE = 0 AFTER 2.5m
 IE = 1 AFTER 10.5mm
 OE = 0 AFTER 1m
 OE = 1 AFTER 17.5m
 RESET = 1 AFTER 1m
 RESET = 0 AFTER 4.5m
 RESET = 1 AFTER 15.5m
 SLEEP FOR 14.5m
 DBUS[0..7] = FLT
ENDIF

END

*ENDSCRIPT

*SCRIPT GENERATOR TEST_DAR_CLOCK

ALIAS ONOFF = \$ON\$/OFF

PIN ONOFF
 PIN CLK = FLT

INT i

IF ONOFF = TRUE THEN
 CLK = FLT
ELSE

FOR i = 0 TO 9 STEP 1
 SLEEP FOR 1m
 CLK = 0
 SLEEP FOR 1m
 CLK = 1
NEXT i

ENDIF

END

*ENDSCRIPT

